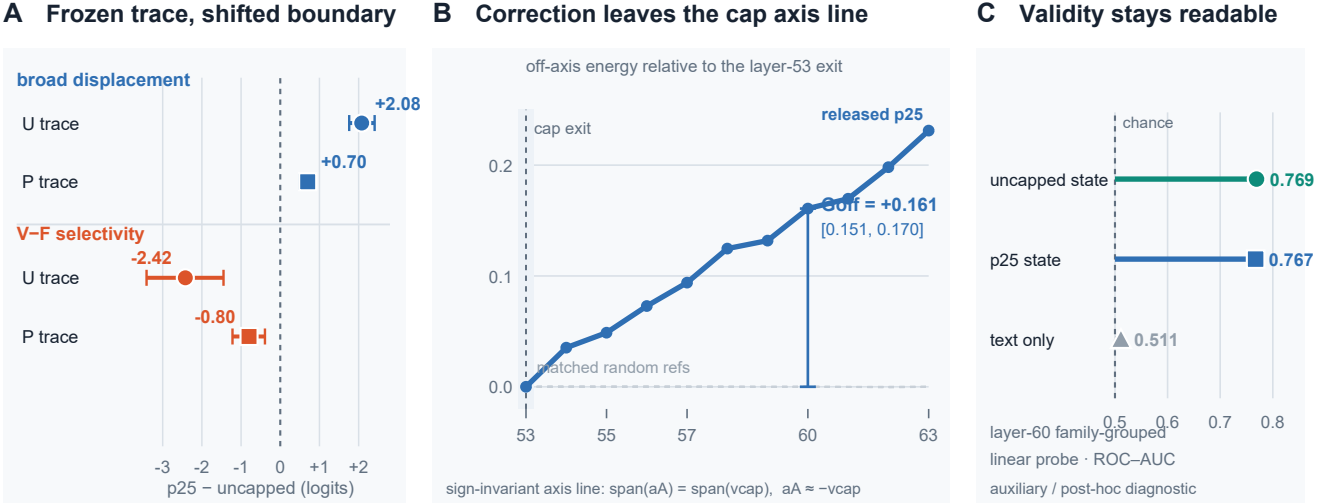

A
Frozen trace, shifted boundary
broad displacement
U trace
+2.08
P trace
+0.70
V−F selectivity
-2.42
U trace
-0.80
P trace
-3
-2
-1
0
+1
+2
p25 − uncapped (logits)
B
Correction leaves the cap axis line
off-axis energy relative to the layer-53 exit
cap exit
released p25
Gain = +0.161
[0.151, 0.170]
matched random refs
0.2
0.1
0.0
53
55
57
60
63
sign-invariant axis line: span(aA) = span(vcap), aA ≈ −vcap
C
Validity stays readable
chance
uncapped state
0.769
p25 state
0.767
text only
0.511
0.5
0.6
0.7
0.8
layer-60 family-grouped
linear probe · ROC–AUC
auxiliary / post-hoc diagnostic